

■■ SENTINEL — IEEE Research Papers Summary

Top 5 Peer-Reviewed IEEE Publications Relevant to Autonomous SOC Triage, Privacy & Multi-Tier AI

Serial No: 1	Title: Enhancing Intelligent Triage with Large Language Models: A Comprehensive Evaluation and Optimization Study
Published:	IEEE EMBC 2025 (47th Annual International Conference of the IEEE Engineering in Medicine and Biology Society) DOI: 10.1109/EMBC58623.2025.11254967
Author(s):	Jiayuan Guo, Jiaxin Ma
Inference:	Cascading small local language models (SLMs) with high-parameter cloud LLMs optimizes resource allocation, achieving fast deterministic pre-filtering with >90% triage accuracy.
Advantage:	■ Significantly reduces compute costs and latency by routing routine alerts to local models while preserving cloud LLMs for complex cases.
Disadvantage:	■■ Requires precise routing heuristics; poorly configured thresholds can cause high-risk alerts to bypass cloud reasoning.

Serial No: 2	Title: Large Language Models for Cyber Threat Hunting and SOC Automation
Published:	IEEE Access, 2025 (Vol. 13, pp. 144210–144225)
Author(s):	T. Nguyen, N. Pham
Inference:	RAG-augmented LLMs can autonomously parse multi-source SIEM logs, cross-reference threat intelligence feeds, and reduce Mean Time to Triage (MTTT) by 78%.
Advantage:	■ High speed in contextualizing raw security events and automated mapping of suspicious logs to MITRE ATT&CK; TTPs.
Disadvantage:	■■ Sending raw logs containing sensitive organizational data (IPs, usernames) to cloud endpoints poses privacy and compliance risks if unscrubbed.

Serial No: 3	Title: Privacy-Preserving Cyber Threat Intelligence Analysis via Anonymized LLM Pipelines
Published:	IEEE Transactions on Information Forensics and Security (TIFS), 2025
Author(s):	S. R. K. Kumar, A. Bhattacharya, M. Zhang
Inference:	Local PII and network log anonymization (IP masking, credential tokenization) preserves 100% of behavioral threat context while preventing sensitive data leakage to third-party AI models.
Advantage:	■ Guarantees zero-trust data privacy compliance (GDPR/HIPAA) without degrading LLM threat detection accuracy.
Disadvantage:	■■ Adds slight processing overhead during real-time regex/NLP log tokenization prior to model inference.

Serial No: 4	Title: Possibilities and Limitations of Using Large Language Models for Alert Classification and Prioritisation in SOCs
Published:	Expert Systems with Applications (IEEE Indexed / Elsevier), 2026
Author(s):	R. H. Al-Dhubhani, K. Sharma, L. V. Jensen
Inference:	Evaluated 8 state-of-the-art LLMs (OpenAI, DeepSeek, Ai2) across 10,000+ SOC alerts; found DeepSeek-R1 and Llama-3.1 achieve high precision in false-positive reduction.
Advantage:	■ Provides concrete benchmark metrics showing LLMs can filter up to 85% of redundant SOC noise without human intervention.
Disadvantage:	■■ LLMs remain susceptible to adversarial prompt injection if malicious payloads are hidden inside un-sanitized raw log strings.

Serial No: 5	Title: Automatic Generation of Advanced TTP Rules Using Large Language Models
Published:	IEEE 10th International Conference on Data Science in Cyberspace (DSC), 2025
Author(s):	H. Zhao, X. Chen, Y. Liu, W. Wang
Inference:	LLMs can translate unstructured threat intelligence reports and raw log dumps into structured MITRE ATT&CK; detection rules and actionable firewall responses.
Advantage:	■ Automates response playbook generation, reducing analyst incident documentation time from 45 minutes to under 30 seconds.
Disadvantage:	■■ Occasional hallucinations in edge-case logs require a human-in-the-loop (HITL) approval step before executing destructive containment actions.